THE

SCARA Robot

THE Series Safety Manual
(Compatible Controller: TS5000-MS)

Rev. 3

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Important

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

This manual is applicable to the following robot.
THE Series: THE600

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Introduction

Thank you very much for your purchase of SHIBAURA MACHINE's robot system.
This manual describes the safety measures for the system robot THE series built by SHIBAURA MACHINE.
Please read through this manual carefully and handle the robot according to the instructions given throughout the manual so that you can completely understand the performance of the robot and use its functions safety over the long years to come.

This manual has the following chapters.

• Use Range
This chapter describes the use range intended by this product.

• Use Method
This chapter describes main specifications and safe use method of this product.

• Warranty
This chapter describes the scope of warranty.

• Standards, laws and regulations regarding safety of industrial robots
This chapter describes main standards, laws and regulations that should be observed to use the robot.

• Safety Precautions
This chapter describes important information on using the robot safely and properly.

• Service Network
This chapter describes technical inquiry and service contact.

• Locations of Warning Labels
This chapter describes the locations of warning labels affixed to the robot and controller.

• Safety Measures
This chapter describes the safety functions of the robot and controller, and safety precautions for installing and operating the robot.

• Applicable Standards and Safety Measures
This chapter describes the standards and safety measures with which the robot and controller are compliant.

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Use Range

This product is the horizontally articulated type industrial robot in which the manipulator has two horizontally rotating joints and the mechanical interface on its tip moves vertically and rotates horizontally.
The product has been developed with intention of adapting it to automation of wide range of work including conveying work such as movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product with the aim to adapt it to automation of these work and incorporate it to the automation equipment.

This product is not intended to apply it to work that contacts a human (such as work related to medical care and nursing care), incorporate it to the equipment that supports human life (such as life support device and incubator), incorporate it to the equipment on which a human rides (including transport equipment such as car, aerospace equipment such as airplane, and game equipment such as roller coaster), and incorporate it to the equipment that protects human life and human body (such as safety device).
This product is not intended to incorporate it to the equipment aimed to control large quantity of heat (such as nuclear power controller and incinerator controller).
This product is prohibited from being used to apply it to work or incorporate it to the equipment that is likely to threaten human life and harm a human body directly due to a failure or malfunction. If this product is used, it will be guaranteed under customer's responsibility.

Use Method

The use conditions for the THE series robot are described. If the product is used in the state that differs from the conditions, unexpected risk may occur. Abide by the conditions for use.

1. Main Specifications of Product: See Specifications "1.4 Specifications Table" and Installation "3.1 Installation Environment" in the Robot Manual.
2. Restrictions on Use: Tables 1 and 2 give "Restricted Specification Sheet" indicating restrictions on use.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer to the installation site. - Controller [1] Use a cart to transfer to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Push the robot by hand to move to a location where it is easy to perform work. [6] Mount the hand. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint axis and the origin position. [8] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power to robot, controller, and optional parts while their covers are removed - Operation in other installation state than the specified (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Entry into the hazard zone of the robot when the servo power is on - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Movable range	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: Repacking: Maintenance: Disposal: 	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m^3 shall be secured. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.

Table 3. Restricted Specifications Sheet

3. Movable range of product (restriction on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - The drive power shall be turned off whenever a person enters the robot hazard zone. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of "transfer and installation" (transportation, unpacking, transfer, installation, and repacking), "use" (teaching, operation, and maintenance), and "disposal."

Warranty

We conduct a strict tests and inspections before delivery to ensure that the product satisfies our performance standards.
We provide warranty to cover defects of the product based on the following prerequisites.

1. Warranty period The warranty period of the product is either of the following, whichever is earlier.
   1) 18 months after delivery from our factory
   2) 12 months after installation at your site

2. Warranty scope
   1) We provide warranty for the product. Note that only the specifications and functions defined in the product specifications, catalog, and instruction manual are covered by the warranty. Any secondary or incidental damages caused by failure of the product are not covered by the warranty in any circumstances.
   2) We provide repair of a faulty product for free only if the product has been handled and used according to the instruction manual that comes with the product.
      Any repairing after the warranty period passes will be charged.

3. Disclaimer The warranty shall not apply to the following cases.
   a) Any failure or damage arising from your careless use or misuse that is not defined in the instruction manual.
   b) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables[*1)].
   c) Any failure regarding sensuous phenomena such as noise not affecting the functions
   d) When any modification or disassembly without our consent is confirmed
   e) Any failure or damage arising from insufficient maintenance and inspection or inappropriate repair
   f) Any failure or damage due to act of God, fires, and other force majeure
   g) Internal data such as programs and points created or modified by users
   h) When a product purchased in Japan is taken abroad

4. Precautions
   1) SHIBAURA MACHINE shall not guarantee the standard performance of the product if you use the product beyond the specifications.
   2) If the customer did not observe the “DANGER,” “WARNING,” and “CAUTION” described in this instruction manual, SHIBAURA MACHINE will not assume the responsibility for any consequential accident resulting in injury or death, damage or trouble.
   3) “DANGER,” “WARNING,” “CAUTION” and other descriptions stipulated in this instruction manual are only those which can be assumed by SHIBAURA MACHINE without predicting all events in all situations. Please no that the range is limited.

(*1) Consumable parts refer to the replacement parts for maintenance as listed in instruction Manual “Robot Manual: Maintenance.”

Standards, laws and regulations regarding safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" defines measures for reducing risks regarding machinery.

Step 1: Fundamental safe design measures - Restrictions on force, speed, and energy

Step 2: Safety protection - Installation of protective fences, etc.
Additional protection measures - Installation of emergency stop devices

Step 3: Information for use - Warning labels, alerts, instruction manuals

Based on the above, international standard ISO/IEC defines “Guide 51” that groups various safety standards in a hierarchical structure.

Safety standards for industrial robots are standardized as separate safety standards for machinery in layer C.

- ISO 10218, ISO 10218-1

After risk assessment is carried out, provision of residual risk information to the user is standardized.

- IEC 82079-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

Safety Precautions

This manual contains important information for the safe use of the robot and the prevention of injury to the operator and others and the prevention of damage to property.
Read and understand the meanings of the following indications and symbols first, and observe these precautions.
To install and operate the robot safely, read and understand this manual in advance.

[Meanings of Indications]

Indication	Meaning
⚠DANGER	Indicates the imminent danger that incorrect operation could cause death or serious injury.
⚠WARNING	Indicates the danger that incorrect operation could cause death or serious injury.
⚠CAUTION	Indicates the danger that incorrect operation could cause injury[1].
NOTICE	Indicates that the property damage[2] may occur.

1) Injury indicates injuries such as wounds, burns, and electric shock that do not require hospitalization or long-term outpatient treatment.
2) Damage to property indicates extensive damage related to the destruction of assets or materials.

[Transportation]

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "THE 600 Robot Manual: Installation" together with this manual to strictly observe the following items.

⚠ CAUTION

- Be sure to convey the product by two or more workers.
  Working in an unnatural posture causes the packing box to drop.
  - Secure the transportation route in advance.
  - Use the cart, etc. to carry the product.
  - Wear the protective shoes.

NOTICE

- Be sure to convey the product with by two or more workers.
  Working in an unnatural posture causes the packing box to drop and the product in the box to be damaged.
  - Do not use the dropped product.

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "TS5000 controller Manual: Installation" together with this manual to strictly observe the following items.

⚠ CAUTION

- Do not stack the packaging boxes on top of each other. The load could topple over, resulting in an injury.
- Do not allow water to splash on the controller. Moisture could get inside the controller, resulting in a breakdown.
- If the controller is carried manually, be sure to wear safety gloves and safety shoes. Otherwise, your fingers, hands, or feet could get trapped by the controller.
- Use a cart or similar device when carrying. The controller could be dropped if carried by hand.

NOTICE

- Do not subject to excessive vibrations during transportation. This could damage the controller.
- Do not transport while upside-down or store in a vertical orientation. This could damage the controller.

[Unpacking]

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "THE 600 Robot Manual: Installation" together with this manual to strictly observe the following items.

⚠ DANGER

- Never use the damaged robot.
  Otherwise, the faulty robot may cause unpredictable disaster.
  - Check the appearance based on the external view described in the manual.
  - Check the parts based on the accessory list attached to the controller.
  - If damage is found or parts are insufficient, please contact SHIBAURA MACHINE Sales Division.

⚠ CAUTION

- When cutting the packing rope with the cutter, never direct the blade toward a person.
  Otherwise, the person may get a cut due to the cutter.
  - Use the nippers or scissors.
  - Wear gloves.
- When putting the robot, cables, etc. in and out of the packing box, never lift them up in an unnatural posture.
  Otherwise, excessive burden may damage worker's back.
  - Secure space where work can be performed easily.
  - Put the robot, cables, etc. out of the packing box by multiple workers according to the situation.

NOTICE

- Never use the damaged robot.
  The robot is faulty.
  - Check the appearance based on the external view described in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or the parts are insufficient, please contact SHIBAURA MACHINE Sales Division.
- Be sure to dispose of the unnecessary packing materials according to the customer's in-house regulations.
  To store the packing materials, be sure to manage them to prevent them from being used for other than packing the robot.

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "TS5000 controller Manual: Installation" together with this manual to strictly observe the following items.

⚠ CAUTION

- Wear protective gear when disposing of the unneeded pallets and cardboard boxes after unpacking. Otherwise, you could be cut by nails sticking out and by other sharp locations, resulting in an injury. Also, dispose of any unneeded wooden pallets, cardboard boxes, plastic bags, cushioning materials, and other items in compliance with the regulations of your company.
- Use a dedicated cardboard cutter when opening the packing box. You could cut your finger or hand if using a cutter with a sharp edge.
- Unpack at a stable location. If unpacking at an unstable location is performed, the controller could fall, and you could get your hands or feet trapped, or you could hurt your back.
- Wear safety gloves when unpacking. Your hand could get trapped between a cardboard box and the controller, resulting in an injury. You could also be injured by the corners of the controller or cardboard box.

NOTICE

- Do not allow water to splash on the controller. Moisture could get inside the controller, resulting in a breakdown.

[Location]

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "THE 600 Robot Manual: Installation" together with this manual to strictly observe the following items.

⚠ CAUTION

- Never convey the robot by one worker.
  Otherwise, the mass exceeding 31 kg cannot be supported, and the robot will drop.
  - Convey the robot by two or more workers.
  - Use the cart, etc.
  - Secure the conveyance route in advance.
- Be sure to convey the product by two or more workers.
  Working in an unnatural posture causes the packing box to drop.
  - Secure the transportation route in advance.
  - Use the cart, etc. to carry the product.
  - Wear the protective shoes.
- When conveying the robot, be sure to use the attached clamp.
  Otherwise, the worker may get caught in the robot arm that moved.
  - Wear gloves to prevent slip and protect hands.
- When lifting up the robot, be sure to lift it up slowly.
  Otherwise, the tilted robot may collide with the worker.
- Use the wire whose limit load can sufficiently bear the weight of the robot.
  Otherwise, the wire may be broken and the robot may drop.

NOTICE

- Never touch the ball screw spline shaft by bare hand.
  Otherwise, rust will develop early.
  - When touching the ball screw spline shaft, wear gloves.
  - If the ball screw spline shaft is touched by bare hand, grease up the ball screw spline shaft.
- After installation is completed, be sure to remove the clamp and eyebolts.
  If the robot is moved with them mounted, the robot may be broken or become faulty.
- Be sure to lift up and lift down the robot silently.
  Otherwise, the robot may collide with the peripheral equipment, frame, etc. to cause failure or damage.

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "TS5000 controller Manual: Installation" together with this manual to strictly observe the following items.

⚠ CAUTION

- Do not carry by holding the AC power supply unit. If the controller is carried by holding the AC power supply unit, it could be damaged and fall. Also, an electric shock could occur if AC power is being supplied.
- If the controller is transported manually, be sure to wear safety gloves and safety shoes. Otherwise, your fingers, hands, or feet could get pinched by the controller.
- Use a cart or similar device when carrying and transporting. The controller could be dropped if carried by hand.
- Transport while the controller cables are removed. If the controller is transported while the cables are still connected, a cable could be broken, or an electric shock could occur.

NOTICE

- Do not subject to excessive vibrations. This could damage the controller.
- Do not allow water to splash on the controller. Moisture could get inside the controller, resulting in a breakdown.

[Installation]

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "THE 600 Robot Manual: Installation" together with this manual to strictly observe the following items.

⚠ DANGER

- Be sure to connect wiring after the robot is installed.
  Otherwise, a worker may stumble over the cable and fall. The worker may touch the cable that has been broken due to a dropped carried object and receive electric shock.
  - House wiring in the duct, etc.
- Be sure to connect a grounding wire.
  Otherwise, the robot that has operated unpredictably due to noise may collide with a worker. Electric leakage may result in electric shock or fire.
- Never install the robot on a frame that has low rigidity or a weak foundation.
  Otherwise, the robot may fall over together with the frame during operation.
  - See the load on the frame during robot operation to prepare the frame.

⚠ CAUTION

- When mounting and removing the clamp, never hold the edge of the clamp.
  Otherwise, if forcible force applies, hand or finger may be cut by the clamp.
  - Wear gloves.

NOTICE

- Never install the robot on a frame that has low rigidity or a weak foundation.
  Otherwise, the robot may vibrate during operation. It may become faulty.
  - See the load on the frame during robot operation to prepare the frame.
- Be sure to connect wiring after installation is completed.
  Otherwise, a carried object may drop on the wired cable, resulting in breakage.
  - House wiring in the duct, etc.
- Be sure to connect a grounding wire.
  Otherwise, the robot may become faulty due to noise.
- Be sure to install the robot on a level place.
  Otherwise, the performance and lifespan of the robot may be reduced. The robot may become faulty.
- Be sure to secure the frame on which the robot is installed to the floor or wall.
  Otherwise, the frame will move due to the force exerted during robot operation.
- The cable between the robot and controller is not a movable cable. If the robot is mounted on the traveling shaft, etc., the cable will be broken early.

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "TS5000 controller Manual: Installation" together with this manual to strictly observe the following items.

⚠DANGER

- Never install in a location where the water exposure exceeds the protection rating. This could result in an electric shock or fire.
- Never put near flammable objects. If sparks occur due to a breakdown, nearby flammable objects could ignite, resulting in a fire.
- Do not place objects on top of the controller. (Stacking prohibited)
  - If there is not adequate space for ventilation, heat could build up inside, resulting in a fire.
  - The controller was designed assuming that objects would not be placed on top of it, and so the top could be unable to withstand the weight and collapse, damaging the controller, and causing your hands or feet to get trapped.
- Never use unauthorized cables. Manufacturing and use of unauthorized cables can result in a fire.
- Never connect to unauthorized connectors or terminals. Wiring of unauthorized connectors and terminals can lead to a breakdown of the controller, electrical shock, or a fire.
- Be sure to always install the controller outside the hazard zone and at a location where robot operations are visible.
  If the controller is operated within the hazard zone or the controller is used at a location where the robot or peripheral devices are not visible, an operator or worker within the hazard zone could collide with a robot or peripheral device.
- If the controller is mounted in an airtight space such as in a control panel, provide air holes, forced ventilation, and indirect cooling so that no heat buildup occurs. Otherwise, if a controller is mounted in an airtight space, heat could build up, resulting in a breakdown or a fire.
- Either mount the controller inside the control panel so that it cannot be contacted by workers, or mount in a location that can provide sufficient distance from workers. If the controller is installed within the movable range of workers, they could get their hands or feet entangled in the cables, causing them to trip, or an electric shock could occur due to contact with current-carrying parts.
- Do not install on an unstable location. The controller could fall, and you could get your hands or feet trapped.
- Connect the robot wiring after installation is complete.
  - If the controller is installed or transported while the wiring is connected, a cable could be broken, resulting in an electric shock.
  - Feet could get entangled in the wiring, resulting in a fall.
- Use a dedicated duct to protect the lead-in cables of the control panel. Use of bare cables can lead to cable breakage, resulting in a fire or electric shock.
- Mount a cover or other enclosure on the buttons and switches on the control device connected to the controller. If a signal is turned on due to unintentional contact, the robot could be activated, resulting in a collision.

⚠ DANGER

- Display the operation mode of each controller on the operation device. If you are unaware of the operation mode, a collision could occur with a robot that is performing an unexpected operation.
- Always connect the ground wire. Usage without connecting the ground wire can result in ground leakage, electric shock, or a fire.
- After connecting, use a tester or other instrument to check the voltage and for short-circuits. If the power is turned on when there is miswiring or during a controller breakdown, an electric shock or fire could result.
- Firmly connect the connectors. If the connectors do not fit firmly, electrostatic charges could build up and lead to a malfunction, which could result in collision with a robot.

⚠ WARNING

- Turn off the power before connecting and disconnecting cables from the controller. Touching the terminal for a cable that was removed while the power is on could result in an electric shock, and connecting and disconnecting cables while the main power is on could damage the ICs in the board.
- When disconnecting cables, grasp the connector while holding the controller, and pull to remove.
  - Pulling the cable itself could break the cable and result in an electric shock.
  - The controller could flip over and be damaged.

NOTICE

- Never supply a current that exceeds the rated output. If a current exceeding the rated output is supplied, the output elements, boards, or other components could be damaged, resulting in a fire.
- If the LAN, OUTPUT, or SD (option support: TRIG, CONV) connectors will not be used, be sure to always mount a connector cap (a dummy SD card for the SD connector). If no protection is provided, a breakdown could occur due to an electrostatic charge.
- To ensure adequate ventilation around the controller, be sure to provide at least 50 mm of clearance on the right, left, and rear sides, at least 100 mm of clearance on the top side, and a clearance equivalent to at least the height of the rubber feet (11 mm) on the bottom side. If the ventilation space is not provided, the controller could overheat and break down.
- If the controller is mounted in an airtight space such as that for a control panel, provide air holes, forced ventilation, and indirect cooling so that no heat buildup occurs. Otherwise, if a controller is mounted in an airtight space, heat could build up, resulting in a breakdown or a fire.
- When mounting in the control panel, use anchoring fixtures and other implements to secure the controller in place. If the control panel is subjected to vibrations, it could move inside and fall, resulting in a fire.
- Set the hand output signal to 0.1 A or less per output (total for 8 outputs is 0.8 A or less). If a current exceeding the rated output is supplied, the output elements, boards, or other components could be damaged, resulting in a fire.
- Apply a voltage in the range of 24 V±10% to the external input/output power cable. Applying a voltage that exceeds the allowable voltage could damage the controller and result in a fire.

[Teaching]

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "TS5000 controller Manual: 5.2Precautions for teaching operations" together with this manual to strictly observe the following items.

⚠ DANGER

- Be sure to perform the operation according to the instruction manual to avoid being struck by a robot or receiving an electric shock.
- Be sure to perform teaching operation from outside the hazardous area to avoid being struck by a robot. Observe the following instructions when teaching in a hazardous area.

(1) The operation should be performed by at least two people. One person does the teaching while other one monitors from outside the hazardous area. Be sure to check on each other to prevent any misoperation.

(2) When teaching in a hazardous area, the operator must hold the controller's master key and the teach pendant to prevent third-party operation.

(3) The operator should work in a way to always be able to push the emergency stop button. After confirming the operation area of the robot and surrounding shields, the operator works at a position where he can retreat immediately in the event of an abnormality.

(4) The supervisor monitors the operation from a safe position where the entire robot can be seen, and where the emergency stop button can immediately be pushed in the event of an abnormality. Also, the supervisor should keep other people away from entering the hazardous area. If an operator or another person does not follow the instructions of the supervisor, an accident may occur.

- If an error occurs when the device is turned on, or if the "READY" LED on the controller is not turned on, be sure to turn off the power to prevent electric shocks or fire.
- Be sure to perform daily inspections before operating the equipment. Failure to do daily inspections increase the risk of an accident.
- When turning on the servomotors, be sure to exit the hazardous area. There is a risk of being struck by a robot that moves unexpectedly.
- If a failure or malfunction occurs, be sure to stop the robot and turn off the controller power to prevent a fire or a crash of the robot.

⚠ WARNING

- Never enter a hazardous area during test operation as you may be hit by a robot.
- Never leave equipment in the work space that could interfere with your work and create a risk of stumbling or falling. When concentrating on the teaching operation, attention to the surroundings is neglected and the risk to crash into peripheral devices is increased.
- Never allow anyone other than the operator to approach the robot to prevent being hit by a robot.

⚠ WARNING

- Never open the controller's cover during operation. Do not connect nor disconnect the connectors to avoid receiving an electric shock. Touching the connectors during operation increases the risk of being hit by a robot moving unexpectedly.
- Never change the data in the system configuration file during operation to avoid abnormal operations or crash of the robot.
- If an abnormal operation is confirmed during a teaching operation, be sure to stop using the robot as it may lead to a more serious accident.
- Be sure to stop operation if there is any abnormal noise, odor, or unexpected behavior from the controller or robot. If you ignore the abnormality and continue to operate, the risk of fire or crash due to abnormal operation will be increased.
- When teaching in the hazardous area, be sure to operate with the teach pendant or to hold the master key. If a robot is started by a third party during operation, there is a risk of being struck by a robot.
- Be sure to turn off the servomotors when leaving the workplace to avoid another operator from being hit by accident by the robot.
- If a failure or malfunction occurs, be sure to stop the robot and turn off the controller power as there is a risk of fire or crash of the robot.
- When making the third axis servo-free, be sure to not put your hands or feet under the third axis. The third axis will fall and injure you by pinching your hands and feet.

⚠ CAUTION

- All hand output signals are turned off when the controller's power is turned off or when the power is turned on. Be sure to design the end effector so as not to drop objects due to a power failure. There is a risk of having the hands and feet pinched by the dropped end effector.

NOTICE

- Be sure to turn off the servomotors' power while the robot is stopped.If the main power is turned off while the robot is operating, the arm may keep moving and collide with peripheral devices.
- When performing automatic operations, be sure to confirm the program execution status as there is a risk to interfere with peripheral equipment and cause the device to malfunction.

[Operation]

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "TS5000 controller Manual:5.3 Precautions for automatic operations" together with this manual to strictly observe the following items.

⚠ DANGER

- Never operate if there are damaged or missing parts as there is a risk of electric shocks or fire.
- Do not put conductive objects or fingers in the power supply terminal area to avoid receiving an electric shock.
- The operation mode of each controller should be displayed on the controller. Not knowing the operation mode increases the risk of being struck by an unexpected robot.
- Be sure to perform the operation according to the instruction manual to avoid being struck by a robot or receiving an electric shock.
- If an error occurs when the device is turned on, or if the "READY" LED on the controller is not turned on, be sure to turn off the power as there is a risk of electric shocks or fire.
- When turning on the servomotors, be sure to exit the hazardous area to avoid being struck by a robot that moves unexpectedly.
- If a failure or malfunction occurs, be sure to stop the robot and turn off the controller power as there is a risk of fire or crash of the robot.

⚠ WARNING

- Never open the controller cover during operation to avoid receiving an electric shock.
- Never change the data in the system configuration file during operation to avoid abnormal operations or crash of the robot.
- When using multiple controllers, be sure to select the same operation mode for all of the robots to avoid being struck by an unexpected robot.
- Do not enter a hazardous area during test operation to avoid being hit by a robot.
- Do not allow anyone other than the operator to approach the device to avoid being hit by a robot.
- Be sure to perform daily inspections before operating the equipment. Failure to do daily inspections increases the risk of an accident.
- Be sure to stop any operation if there is any abnormal noise, odor, or unexpected behavior from the controller or robot. If you ignore the abnormality and continue to operate, the risk of fire or crash due to abnormal operation will be increased.
- Do not enter the dangerous area of the robot during automatic operation to avoid being struck by a robot.
- Be sure to check the safety protection devices before starting any automatic operation. If the safety protection devices break down, it will not be able to perform an emergency stop, leading to the collision with the robot.

NOTICE

- Do not put obstacles within the robot movement range to avoid collision.
- The hand output signal should be 0.1A or less per signal (total of 8 signals equals to 0.8A or less). If a current exceeding the rated output is applied, the output element (board, etc.) may be damaged and cause a fire.
- The external I/O power supply should be of 24V±10% (24VDC). Applying a voltage exceeding the allowable value may damage the controller and cause a fire.
- Be sure to turn off the servomotors power while the robot is stopped.
  If the main power is turned off while the robot is operating, the arm may keep moving and collide with peripheral devices.
- When performing automatic operations, be sure to confirm the program execution status as there is a risk to interfere with peripheral equipment and cause the device to malfunction

[Maintenance]

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "THE 600 Robot Manual: Maintenance Manual" together with this manual to strictly observe the following items.

⚠ DANGER

- Never replace and modify the parts for other than permitted portions.
  Otherwise, the faulty robot may cause unpredictable disaster.
- When opening the cover, never allow entry of moisture or contaminant into the robot.
  If the power is turned on while moisture or contaminant is left, an electric shock may result.
- Never touch liquid that leaked from the battery.
  If liquid adheres to your skin or eye, inflammation results.
  - When liquid adheres to your skin, wash it away with flowing water immediately.
  - When liquid adheres to your eye, wash it away with flowing water immediately and then see a doctor.
- Replace the parts to which liquid has adhered.
- When performing maintenance with the power turned ON in the dangerous range of the robot, be sure to switch the master mode key to "MANUAL" to pull out the key and carry it with the teach pendant.
  Otherwise, the robot that has operated unpredictably may collide with a worker.
  - The worker should receive safety training regarding industrial robot inspection-related work defined by each country's laws and regulations.
  - Display the fact that work is in progress near the controller.
- When performing maintenance with the power turned ON in the dangerous range of the robot, be sure to arrange a work observer outside the dangerous range of the robot.
  Otherwise, when a worker in the dangerous range of the robot suffers damage, the worker cannot be rescued.
  - The observer should secure safety of the worker and the third party who approaches the dangerous range of the robot.
  - The observer should also receive safety training regarding industrial robot inspection-related work defined by each country's laws and regulations.
- When approaching the robot being stopped, be sure to check the robot operation status.
  If the robot is under operation, the robot that has restarted may collide with the worker.
  - Check the operation status indicated on the indicator lamp of the controller and on the teach pendant.
  - The robot operation status is output by signals (servo operation preparation completion, external mode, operation preparation completion, automatic operation, failure, and emergency stop). Connect to the display equipment, etc.
- Be sure to perform maintenance and inspection periodically.
  If maintenance and inspection are neglected, the faulty robot may cause an unpredictable disaster.

⚠DANGER

- When replacing a motor, be sure to remove the power plug of the controller.
  Otherwise, the removed cable terminal may be touched to result in an electric shock. The worker may be entangled in rotation of the motor that has made unpredictable motion.
- When replacing the timing belt, be sure to remove the power plug of the controller.
  The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.
- When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
  The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.
- When replacing the reduction gear, be sure to remove the power plug of the controller.
  The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.
- When replacing the LED lamp, be sure to remove the power plug of the controller.
  The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.
- When performing maintenance with the power turned on in the dangerous range of the robot, be sure to switch the master mode key inserted to the teach pendant to “MANUAL” (for TP5000) to pull out the key and carry it with the teach pendant.
  Otherwise, the robot that has operated unpredictably may collide with a worker.
  - The worker should receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.
  - Display the fact that work is in progress near the controller.
- When approaching the robot being stopped, be sure to check the robot operation status.
  If the robot is under operation, the robot that has restarted operation may collide with the worker.
  - Check the operation status indicated on the indicator lamp of the controller and on the teach pendant.
  - The robot operation status is output by signals (servo operation preparation completion, external mode, operation preparation completion, automatic operation, failure and emergency stop). Connect to the display equipment, etc.
- When performing maintenance with the power turned on in the dangerous range of the robot, be sure to arrange a work observer outside the dangerous range of the robot.
  Otherwise, when a worker in the dangerous range of the robot suffers damage, the worker cannot be rescued.
  - The observer should secure safety of the worker and the third party who approaches the dangerous range of the robot.
  - The observer should also receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.

⚠ DANGER

- When replacing the battery with the power kept on, be sure to emergency-stop the robot.
  Otherwise, the robot that has operated unpredictably will collide with a worker.
  - The worker should receive special training, etc. regarding industrial robot inspection-related work defined by each country's laws and regulations.
- Be sure to replace the parts with SHIBAURA MACHINE-specified replacement parts for maintenance.
  If parts other than the specified parts are used, unpredictable disaster will occur.
  - Items other than the position detector battery are all special-specification products. When purchasing or ordering them, be sure to contact SHIBAURA MACHINE.
- Be sure to contact SHIBAURA MACHINE service company about work other than replacement of the position detector battery. We accepts the work at an extra cost.
  SHIBAURA MACHINE's warranty does not cover accidents and damage resulting from the replacement performed by the customer and its consequence.

⚠ WARNING

- When rotating axis 4 by hand, never insert your hand or finger between the upper or lower mechanical stopper and the robot or under the robot hand.
  Otherwise, your hand or finger may get caught in the mechanical stopper and robot hand.
  - When axis 4 rotates, the ball screw spline shaft will move up and down.
  - The axis-4 motor is not equipped with the brake. If the robot hand is touched by hand when the servo power is off, axis 4 will rotate.
- When releasing the motor brake, be sure to support the ball screw spline shaft.
  Otherwise, the shaft will drop by its own weight concurrently with release of the brake.
  - When a worker supports the shaft, be sure to prevent hand or finger from getting caught.
  - When something strange is felt during work, release the brake release switch immediately. The brake will also be applied by turning off the power to the controller.
  - The brake release switch is located on the robot base rear. Do not press the switch by mistake.
  - The worker should receive safety training regarding industrial robot inspection-related work defined by each country's laws and restrictions.
- When the axis-3 motor is replaced, be sure to lower the ball screw spline shaft to the lower limit.
  When the motor or timing belt is removed, the shaft that drops by its own weight could catch your hand or fingers.
  - When the ball screw spline or axis-4 timing belt or reduction gear is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

⚠ CAUTION

- Be sure to start maintenance when the surface temperature becomes less than 50°C after power off.
  If hot motor or periphery of the motor is touched, burn injury may be received.
- When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
  If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

- Never replace and modify the parts for other than the permitted portions.
  Otherwise, the performance and lifespan of the robot may be reduced, and the robot may malfunction.
- When opening the cover, never allow entry of moisture or contaminant into the robot.
  If the power is turned on while moisture or contaminant is left, the robot may malfunction.
- When mounting the covers, prevent the cables from being caught.
  Otherwise, the cables will be broken.
- Never cause heavy impact on the motor shaft.
  The motor and position detector may malfunction due to impact.
- Never disassemble the motor and position detector.
  Otherwise, they may be disabled due to difference in position or intrusion of foreign material.
- Never forget to fit the O ring, and never use the cut or damaged O ring.
  Otherwise, grease will leak from the fitting surface.
- Never pull out the ball screw nut from the ball screw spline shaft.
  Otherwise, the ball screw spline unit from which balls have come out cannot be used.
- Never touch the ball screw spline unit with bare hands.
  Touching it with bare hands may cause the ball screw spline unit to rust.

- Be sure to wear gloves. • Be sure to perform maintenance and inspection periodically.
  If maintenance and inspection are neglected, the performance and lifespan of the robot may be reduced and the robot may malfunction.

- Be sure to restore where to route the cables pulled out by removing the cover and where to secure with the cable straps to their original state.
  If the cables are pushed by force or where to secure changes, the cables will be worn out or broken.
  - Before pulling out the cables, take pictures for restoration
- When replacing a motor, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.
- After the motor is replaced, be sure to re-set zero position.
  Replacing a motor causes deviation from mechanical zero position; therefore, the robot cannot make normal motion.

NOTICE

- When replacing the timing belt, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.
- After the timing belt is replaced, be sure to re-set the zero position.
  Because a mechanical zero position is shifted with replacement, the robot cannot perform normal operation.
- When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.
- After the ball screw spline unit is replaced, be sure to re-set the zero position.
  Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- When greasing up the ball screw spline unit and apply corrosion inhibitor, be sure to cover the peripheral equipment, etc.
  Grease and corrosion inhibitor may drip on the peripheral equipment.
  - Wipe out excess grease and corrosion inhibitor with waste cloth.
- Be sure to check the application state of grease daily.
  A shortage of grease and development of rust may cause flaws in the sliding surface, resulting in a reduction in the performance and lifespan of the robot. The robot may malfunction.
- Be sure to use the specified type of grease.
  Otherwise, the performance and lifespan of the robot may be reduced. The robot may malfunction.
- Be sure to check the application state of corrosion inhibitor daily.
  Otherwise, rust may develop on the ball screw spline unit.
  - Corrosion inhibitor must be applied only when it is used up.
- Be sure to handle the ball screw spline unit carefully.
  Drop impact or application of excessive external force could ruin the unit.
- After the ball screw spline unit is replaced, be sure to re-set the zero position.
  Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- When replacing the reduction gear, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.
- After the reduction gear is replaced, be sure to re-set the zero position.
  Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- Be sure to handle the arm carefully.
  Drop impact or application of excessive external force could ruin the arm.
- Be sure to handle the reduction gear carefully.
  Drop impact or application of excessive external force could ruin the reduction gear.
- Be sure to use elliptical cam provided for new reduction gear to be replaced.
  The combination of the reduction gear and elliptical cam is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced.

NOTICE

- Be sure to use the specified types of grease.
  Using grease of other than specified types causes malfunction due to poor lubrication.
- Be sure to observe the filling amount of grease.
  If the filling amount is tool large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the filling amount is too small, malfunction is caused by poor lubrication.
- When replacing the LED lamp, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.
- If zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of the teaching point.
  The zero position shifts slightly due to error in setting. If the zero position is used as is, the robot will collide with the peripheral equipment.
- Be sure to replace the position detector battery during annual inspection.
  If position detector data is lost, the robot cannot be operated.
  - When the battery voltage has dropped, a “battery alarm” will occur. At this stage, data is not lost. If the batteries are replaced, the alarm is automatically released.
  - When the battery voltage has dropped further, a “battery error” will occur. At this stage, data is lost. A position detector error alarm will also occur.
- When the robot is not used with the power turned off, be sure to replace the position detector battery every two (2) months.
  Because the battery consumption becomes large in the power off condition, position detector data will be lost before periodic inspection.
- If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.
  The axis 3 operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
  3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
  - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.
- Be sure to enter data to HOME1 and HOME2 for the REORG function when installing the robot.
  Even if data is entered after the zero point has shifted, it cannot be used as data for restoring the zero point.
- Be sure to replace the parts with SHIBAURA MACHINE-specified replacement parts for maintenance.
  If parts other than the specified parts are used, the performance and lifespan of the robot will be reduced. The robot will malfunction.

NOTICE

- The replacement of the axis-4 timing belt requires the axis-3 motor and timing belt to be removed. Therefore, the precautions for the replacement of the axis-3 motor and timing belt must also be observed.
- The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
  Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "TS5000 controller Manual: Maintenance Manual" together with this manual to strictly observe the following items.

⚠ DANGER

- Never perform any part replacement or modifications other than those described in the instruction manual. Reduced performance or a breakdown could occur, leading to an electric shock, fire, or collision due to an unexpected robot operation.
- When opening the cover of the controller, do not allow water or other fluids or foreign objects to get inside. This could cause a short-circuit inside the controller, which could lead to an electric shock or fire.
- Do not attempt to incinerate, disassemble, short-circuit, or charge the battery. This could cause the battery to rupture, resulting in a fire. Dispose of the battery in compliance with the regulations of your company.
- Use SHIBAURA MACHINE special products for replacing certain maintenance parts. Wiring of unauthorized connectors and terminals can lead to a breakdown of the controller, electrical shock, or a fire.
- Be sure to request the SHIBAURA MACHINE servicing company if any maintenance or replacement work not described in this instruction manual is required. (A fee will be charged for this work.) SHIBAURA MACHINE will not be liable for any accidents or damage that occurs due to replacement work performed by the customer or its results.
- Install the switching power supply in the location shown in the instruction manual. If the up and bottom or right and left sides are backwards, or the connector is different, this could cause a short-circuit or broken wire, resulting in an electric shock or fire.
- Always replace with SHIBAURA MACHINE authorized maintenance replacement parts.
  Use of unauthorized parts could result in a fire or other unexpected accident.
- All parts except for the battery for the position detector are special specifications parts. Please request these parts from SHIBAURA MACHINE when purchasing and ordering.
- Except for replacement of button cell batteries and fuses, please request the replacement work from the SHIBAURA MACHINE servicing company. A fee will be charged for this work.SHIBAURA MACHINE will not be liable for any accidents or damage that occurs due to replacement work performed by the customer or its results.

⚠ WARNING

- Do not touch the controller terminals or internal board with your hands for about two minutes after the power cable is disconnected. A residual voltage is still in the large-capacity capacitor inside the controller, and this could result in an electric shock.
- Perform periodic maintenance and inspection. Failure to perform maintenance and inspection could result in controller breakdown or accident.
- Use SHIBAURA MACHINE authorized spare parts when replacing parts. Use of unauthorized parts could result in a fire or a collision due to an unexpected robot operation.
- Before starting any work, be sure to always wear protective gear, and unplug the power cable. If the power is turned on, the power supply board, servo board, switching power supply, and other components will carry a charge, and an electric shock could result.
- Turn off the power before opening the controller cover. Opening the cover while the power is on could result in an electric shock.
- Perform the following actions whenever entering the movable range of the robot.
  - Wear protective gear
  - Place the teach pendant in an accessible location to the worker.
  - Change to MANUAL mode (remove the mode selector key so that the mode cannot be changed)

  ⇒ If the above operations are not performed, the emergency stop could be cleared by another person during the daily inspection work within the movable range, and a collision could occur with a robot that was set to continuous operation.
- Use voltage measurement and the LEDs to check that no PN voltage remains in the capacitor. If a PN voltage remains, an electric shock could occur.
- Do not use a metal scale while the power is turned on. Touching the power-carrying part could result in an electric shock.

NOTICE

- Always replace with SHIBAURA MACHINE authorized maintenance replacement parts.Use of unauthorized parts could result in reduced robot performance and lifespan. A breakdown could also occur.
- Files in the TS5KBKUP folder on the SD card will be overwritten. To avoid overwriting, remove the target files from the TS5KBKUP folder and save it.
- USER BACKUP LOAD deletes all files in the controller. Be sure to first save the files inside the controller on an external device using file transfer.

[Disposal]

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "THE 600 Robot Manual: Maintenance Manual" together with this manual to strictly observe the following items.

⚠DANGER

- Never burn, short-circuit, charge, disassemble, and heat the battery.
  If liquid that leaks after the battery blows out adheres to your skin or eye, inflammation results.
  - When liquid adheres to your skin, wash it away with flowing water immediately.
  - When liquid adheres to your eye, wash it away with flowing water immediately and then see a doctor.
  - Dispose of the battery according to the customer-defined rules.

NOTICE

- Be sure to dispose of the packing materials that have become unnecessary according to the customer-defined rules.
  When storing the packing materials, be sure to manage them to prevent them from being used for other than packing the robot.

Safety precautions on transportation and installation are described. Refer to the Instruction Manual "TS5000 controller Manual: Maintenance Manual" together with this manual to strictly observe the following items.

⚠DANGER

- Never attempt to incinerate, short-circuit, charge, disassemble, or heat the batteries.
  They could rupture and leak, and if the battery fluid gets on your skin or eyes, it will cause inflammation.
  - If battery fluid gets on your skin, immediately flush with water to wash off.
  - If battery fluid gets in your eyes, immediately flush with water to wash off, and then seek medical attention.
  - Dispose of the batteries in compliance with the regulations of your company.

NOTICE

- Dispose of any unneeded packaging materials in compliance with the regulations of your company.When storing, manage the materials so that they are not used for purposes other than packing the robot.

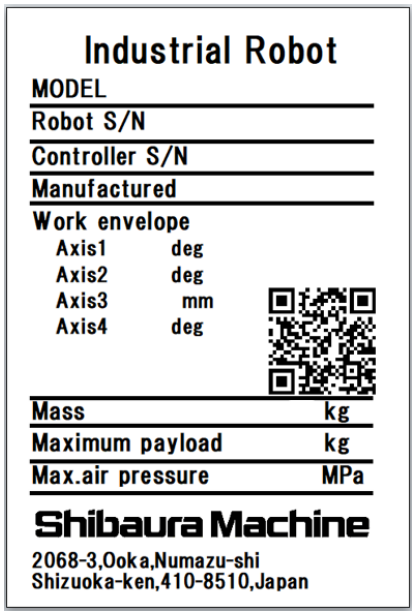
Service Network

Please refer to our distributor's handouts.

1. Warning Labels

1.1 Warning indication contents

The entry contents of the warning labels affixed to the robot and controller are described.

Label layout	Entry contents
	MODEL: Robot model Robot S/N: Robot serial number Controller S/N: Controller serial number Manufactured: Month of manufacture. Year of manufacture Work envelope: Maximum movable range of each joint axis MASS: Robot weight Maximum payload: Maximum payload Max.air pressure: Maximum air pressure * QR code: This scans the model, serial number, and production order number.

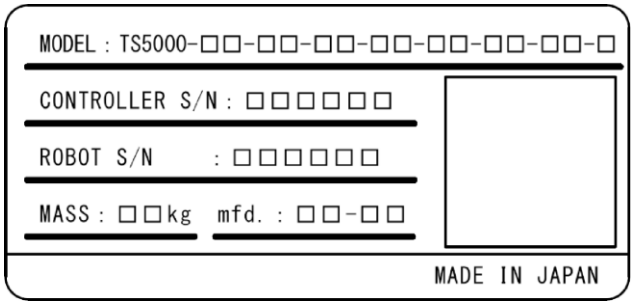
- MODEL, SER.NO., and Mfd.DATE are information required for receiving after-sale service. When calling us at the after-sale service agent, notify the service representative of these information.
- During installation, transportation, and disposal, be sure to confirm the mass before carrying the robot. Increase the number of persons and use the transport equipment as required. The transport equipment should be operated by an operator who has received training pertaining to the transport equipment.

Label layout	Description
	- Keep your hands away from the operating parts of the robot. Your fingers could get entangled, resulting in an injury.
- If you put your hand near the operating parts of the robot, it could get caught in the robot arm or shaft, resulting in an injury. During transfer for installation, transportation, and disposal, secure the arm and shaft in place using the supplied fixture so that they do not move. If you must enter the moving range of the robot to perform teaching or maintenance work for unavoidable reasons, the work must be performed by a trained worker. Install a protective fence to prevent people from entering (entering the moving range of the robot or having a body part get near the operating parts) during operation.	

Label layout	Description
	- Do not enter into the robot's movable range. Otherwise, an injury could result.
- If you enter the movable range of the robot, you could be killed or injured due to contact with the moving robot. If you must enter the moving range of the robot for unavoidable reasons to perform teaching or maintenance work, this must be performed by a trained operator. Implement measures so that no one other than workers performing teaching or maintenance can enter the movable range.	

Label layout	Description
	-High temperature This is affixed to locations that could become hot when operating the robot. Touching these locations could result in a burn. Wait until these high-temperature locations have cooled before performing any maintenance, inspection, or adjustment work.
Wait until these high-temperature locations are fully cooled before performing any work where the robot is touched. If necessary, protect your hands using thermal-resistant gloves or similar protection when performing the work.	

Label layout	Description
	- Be sure to always turn off the controller before performing any maintenance or inspection work. Otherwise, death or serious injury could result due to an electric shock or unexpected movement of the robot.
- If maintenance or inspection work will be performed without turning off the primary power of the controller, the worker could accidentally touch an energized part or charged part, resulting in an electric shock. Also, the robot could start to move due to an unexpected command issued from an external device, contacting the worker and resulting in death or serious injury. Before performing any maintenance or inspection work, be sure to always turn off the primary power, and disconnect the power connector. Also, be sure to post signage that work is in progress so that the primary power will not be turned on during the work. If you must turn on the power to perform work for unavoidable reasons, the work must be performed by a trained worker.	

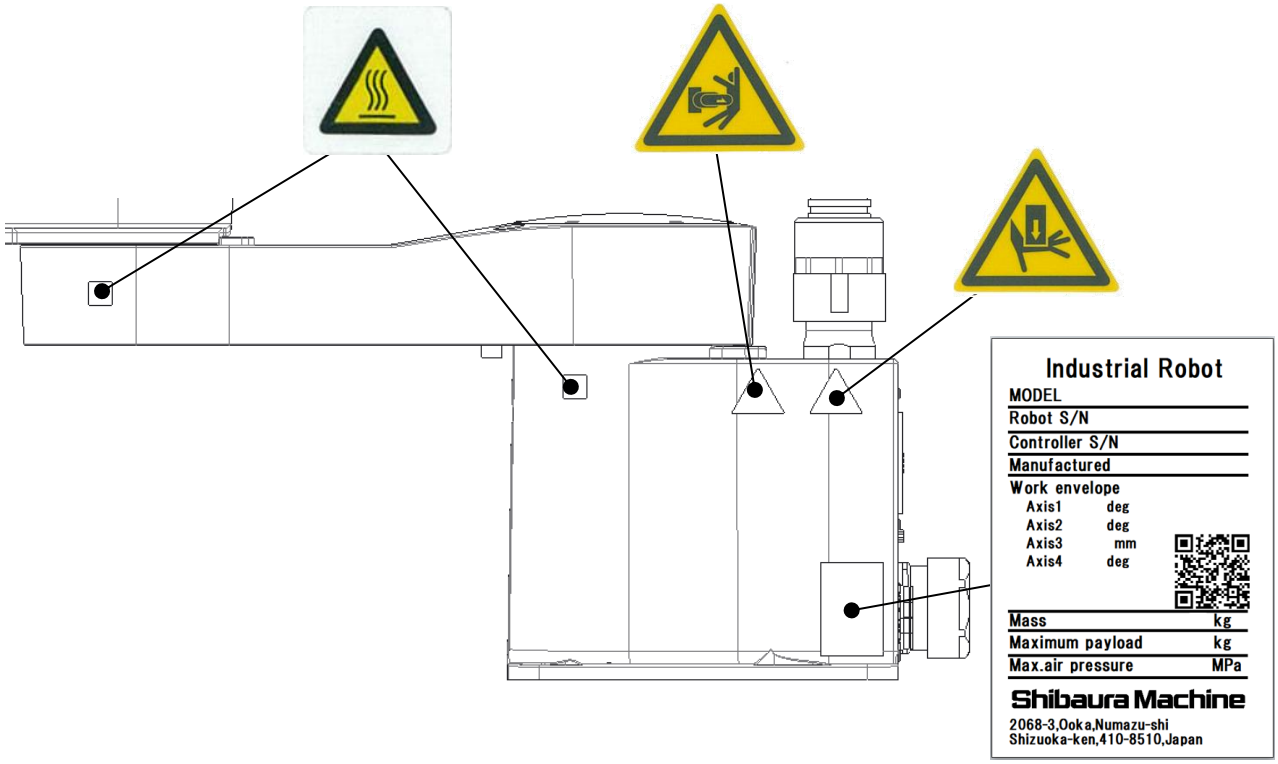
Label layout	Entry contents
	・MODEL：controller model ・CONTROLLE S/N：serial number ・ROBOT S/N：serial number ・MASS：controller mass ・mfd. ：Manufacture date * QR code: MODEL, SER.NO., and manufacturing instruction number can be read.
• If the robot is operated in combination with wrong serial numbers, a phenomenon such as coordinate shift occurs because the individual parameters of the robot do not match. Confirm the serial numbers of the robot and controller and connect and install them in a correct combination.	

Label layout	Entry contents
	• Use the robot body and controller in combination with the correct serial numbers entered on the label. • If an abnormality occurs, turn off the power to the controller, and then contact our service department. • Do not modify this robot and controller without our permission.
• If the robot is operated in combination with wrong serial numbers, a phenomenon such as coordinate shift occurs because the individual parameters of the robot do not match. Confirm the serial numbers of the robot and controller and connect and install them in a correct combination. • If the robot and controller are modified without our permission, they may cause failure or runaway. Do not modify them without our permission. For technical inquiry, call us at the after-sale service agent.	

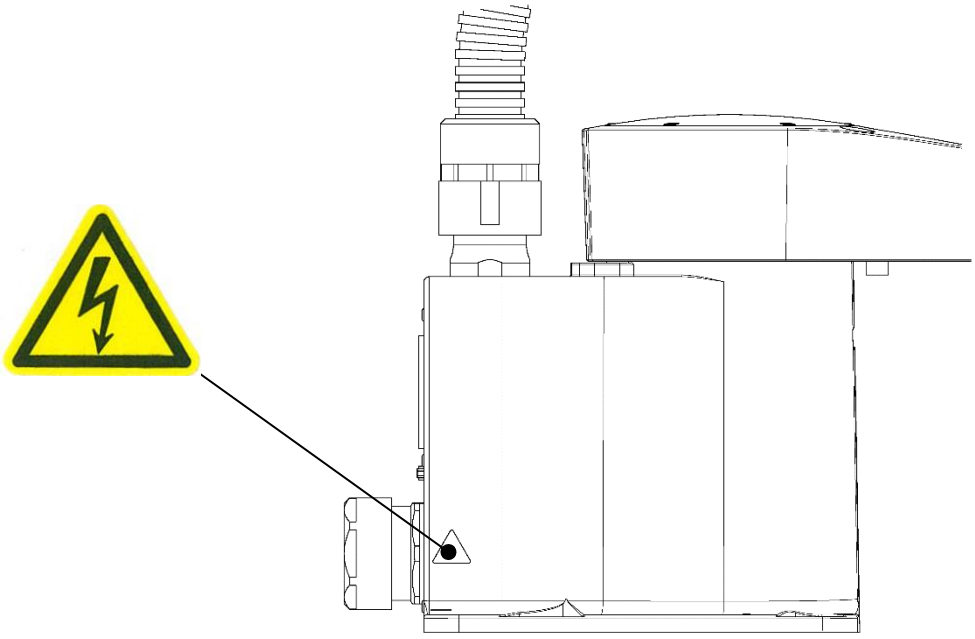
Label layout	Entry contents
	• If an abnormality occurs, turn off the power to the controller, and then contact SHIBAURA MACHINE. • Do not modify this robot and controller without our permission. • Be sure to connect power wire and ground wire of the connector (ACIN) as shown in the following diagram. Mis-wiring causes an electric shock or equipment destruction. • An electric shock may result in death or injury. Before maintenance or inspection, be sure to turn off the power and remove the power connector (ACIN) of the controller. • Do not touch heatsink when power is ON and for a while after power turned OFF. May cause burn.

- If the robot and controller are modified without our permission, they may cause failure or runaway.
  Do not modify them without our permission. For technical inquiry, call us at the after-sale service agent.
- Mis-wiring the power wire and ground wire for use causes an electric shock or equipment destruction.
  Before connecting them, use a tester to confirm whether they are correctly wired. Also, confirm that the primary power supply voltage is within the use range.
- If maintenance or inspection is performed without turning off the primary power to the controller, the conductive or charging portion may be touched by mistake, resulting in an electric shock to cause death or injury.
  When performing maintenance or inspection, be sure to turn off the primary power and remove the power connector. Also, post a notice indicating that operation is in progress to prevent the primary power from being turned on during operation.
  When having to perform operation with the power turned on, a trained operator should always perform the operation.
- During installation, transportation, and disposal, be sure to confirm the mass before carrying the robot.
  Increase the number of persons and use the transport equipment as required. The transport equipment should be operated by an operator who has received training pertaining to the transport equipment.

1.2 Locations of robot warning labels (for THE600)

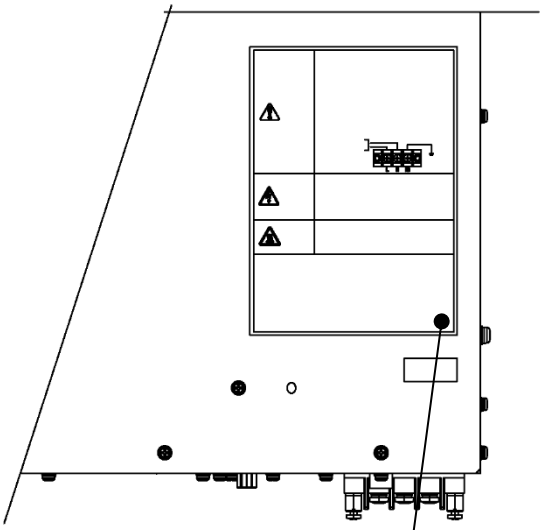


Robot right side

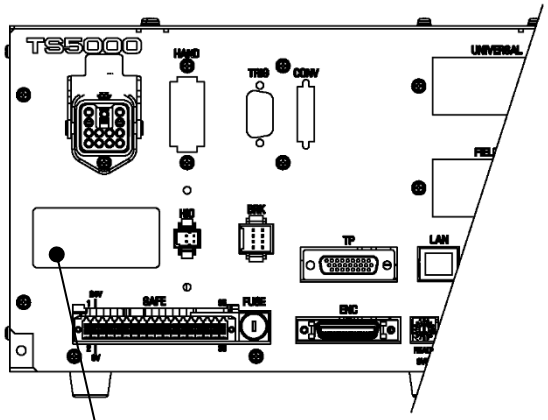


Base rear side

1.3 Locations of controller warning labels (for TS5000-MS)



Top view of controller



Right side view of controller

注 意 CAUTION	■異常の場合はコントローラの電源を切り、弊社までご連絡ください。 In case of trouble, turn off power and call us. ■本ロボットおよびコントローラを弊社に許可なく改造しないでください。 Do not modify this robot and controller without our permission. ■コネクタ(ACIN)の電源線とアース線の接続は必ず下図の配線にしてください。 誤配線すると感電や機器破損の原因となります。 Connect wires for ground and power lines of ACIN as follows. Hazardous voltage may result in electric shock in case of miss-wiring. <CONTROLLER> 1φ AC190～240V 50/60Hz L N PE
警 告 WARNING	■感電により死傷のおそれがあります。 Unit powered by hazardous voltage will cause shock, burn or death. ■保守・点検作業の前に必ず電源を切り、コントローラの電源端子(ACIN)を外してください。 Turn off power and disconnect power terminal(ACIN) before maintenance.
注 意 CAUTION	■電源ON中及びOFFして暫くはヒートシンクに触るな。 火傷の恐れあり。 Do not touch heatsink when power is ON and for a while after power turned OFF. May cause burn.

製造者：
Manufacturer：

住　所：静岡県沼津市大岡2068-3
Address：2068-3, OOKA, NUMAZU-SHI, SHIZUOKA-KEN 410-8510, JAPAN

Robot Controller　　MADE IN JAPAN

MODEL：TS5000-□□-□□-□□-□□-□□-□□-□□-□□-□

CONTROLLER S/N：□□□□□□

ROBOT S/N　　：□□□□□□

MASS：□□kg　mfd.：□□-□□

MADE IN JAPAN

2. Safety Measures

This section describes the necessity of safety measures, safety functions provided for the robot and robot controller, and general safety measures to be taken.

2.1 Cautions on safety

Our SCARA robot and robot controller are equipped with various safety functions.
When the robot is actually operated, however, the following dangers will be supposed.

a) Danger supposed in normal automatic operation

- Operator's mis-operation and mis-judgment, and incomplete program.
- Unexpected robot movement and release or drop of workpiece due to fault of an electronic control device.

b) Danger supposed at teaching and inspection

- Danger of an operator entering the movable range of the robot.
- Operator's negligence from confusion or experience at generation of an unexpected abnormality, and operator's mis-operation due to shortage of behavior and knowledge.
- Approach of an operator to the robot due to unexpected complex movement of the robot.
- Abnormal movement, etc. caused by mis-wiring, poor contact, deterioration and noise

c) Danger supposed in a related machine, etc.

- Sudden movement of the robot with a command from a related machine, etc.
- Sudden movement of a related machine after the robot movement.
- Danger of an operator being caught or entangled in the robot when teaching, inspecting or adjusting the robot while moving a related machine.

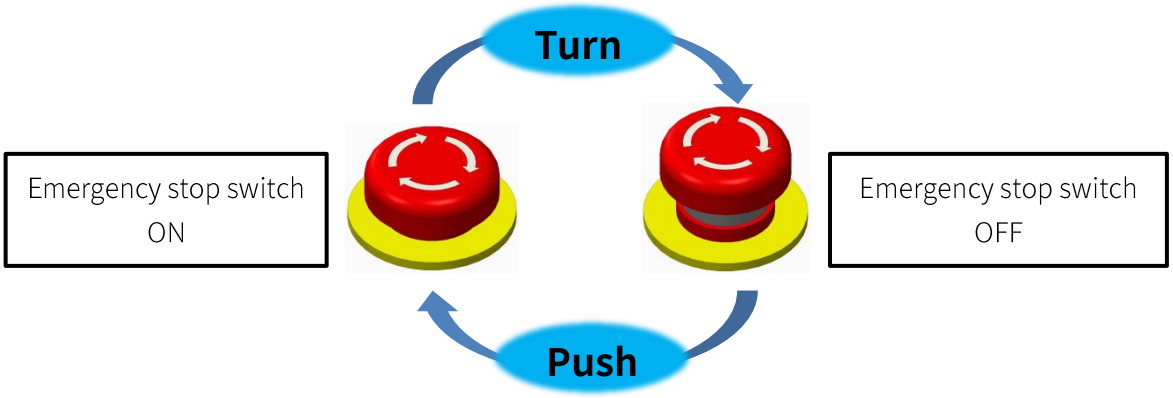
To use the robot safely, safety measures should be taken according to the operating conditions.
Otherwise, an unexpected disaster may occur.
If there are safety rules and regulations, strictly observe the rules and regulations in each country. Also, refer to all manuals relating to this robot and robot controller.

2.2 Safety functions

This robot and robot controller are equipped with various safety functions and additional protection functions.

2.2.1 Emergency stop function

An emergency stop pushbutton switch is provided on the teach pendant. This switch can stop the robot immediately and shut down the robot's power source. (Stop category 1) The emergency stop function is always enabled regardless of the operating condition of the robot.
The emergency stop pushbutton switch is equipped with a lock mechanism. To reset the emergency stop state, pull up the upper part of the switch while turning it as shown in the following figure.
The controller also has the external emergency stop contacts, so an emergency stop switch can be added as required. Select ISO 13850-compliant emergency stop switches as the additional ones.



Precautions for the emergency stop switch
Note the following to select an emergency stop pushbutton switch.

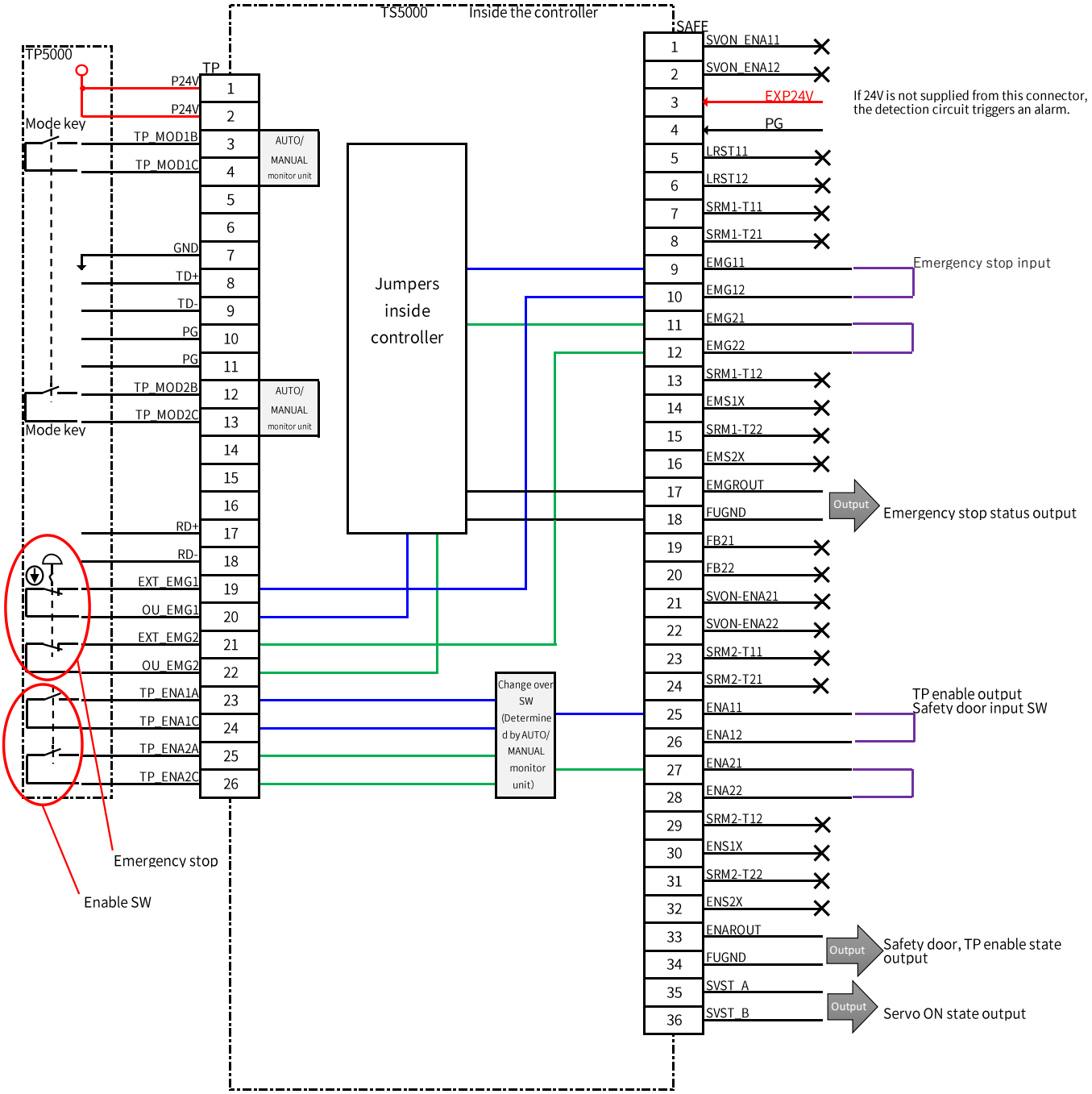
- An emergency stop device must be installed at a location that requires emergency stop and is easy to approach.
- An emergency stop device must be a device that, after functioning, does not allows the circuit to automatically recover until restored manually, and that has a structure where the contact does not recover until the device is reset.
- An emergency stop device must has an appropriate shape like those listed below so that it can eliminate danger according to the structure and characteristics of the machine.
    1) Mushroom-shaped pushbutton
    2) Cable pull switch, lever switch
    3) Switch operated by belly or knee
    4) Pedant switch with no protection cover
- The actuator of an emergency stop device with a pushbutton switch must be red, and the background color of the surrounding must be yellow.
- The cable of an emergency stop device with a cable pull switch must always have a proper tension, and must be distinguished by color (red and yellow).

2.2.2 Safety contact input function

TS5000-MS have the duplicated safety contact input function that is a different system from emergency stop. Connect interlock switches, such as a limit switch and a foot switch, which operate simultaneously with a door of a safety fence.

External emergency stop contacts of TSL3000 are shown in the figure below.

On the TS5000-MS, connect the emergency stop button to the SAFE connector's pins 9-10 and 11-12 and the the safety door to the pins 25-26 and 27-28.

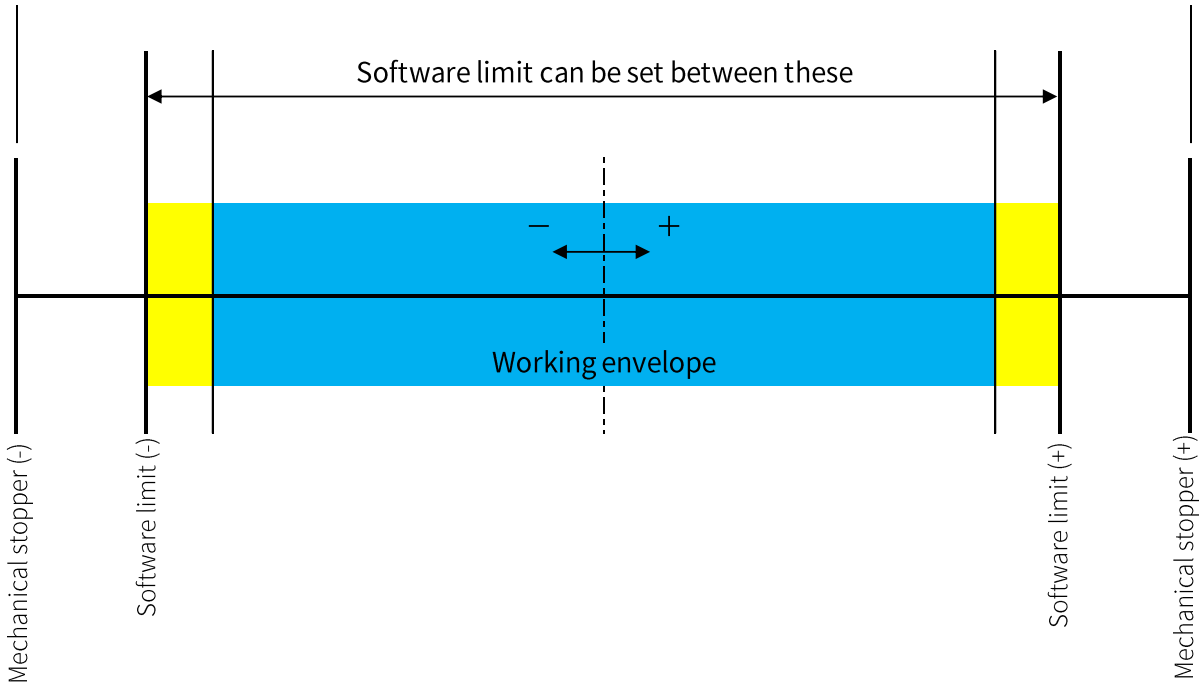


2.2.3 Working range limiting function

This robot is equipped with mechanical stoppers to prevent overrun.
The mechanical stoppers are provided on axis 1, axis 2, and axis 3 in order to prevent the movable part of the robot from overrunning.
The robot working range can be limited by changing the mechanical stopper positions, depending on the robot model.
For change of the mechanical stopper positions, refer to the Instruction Manual "THE Series Robot Manual: Installation."

Software limits can be set as the controller side functions. The software limits are the auxiliary functions of the mechanical stoppers that prevent collision with the mechanical stoppers due to a mis-operation during manual guide or program error. The software limits should be set inside the robot working range limited by the mechanical stoppers.
For change of software limits, refer to the Instruction Manual "TP5000 Operator's Manual."

When the robot operates abnormally, it is unlikely that the software limits are able to limit the working range.
The safety functions aimed to protect human body should be performed with the mechanical stoppers.



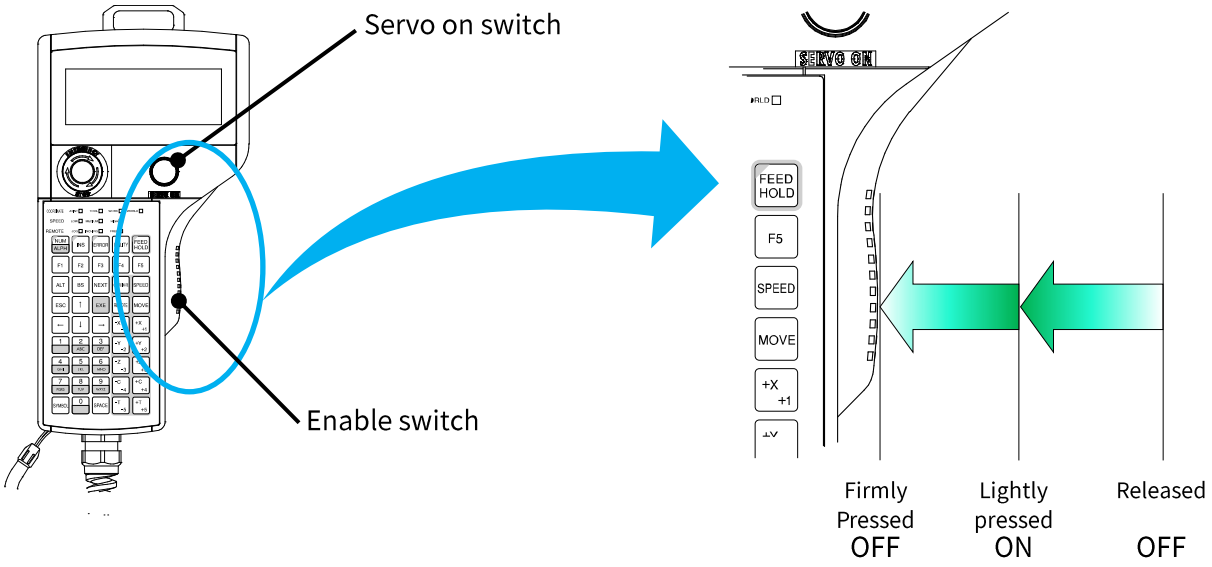
2.2.4 Operation switch on teach pendant

An enable switch is provided on the teach pendant in order to enhance the safety. When guiding the robot in manual mode, the operator holds the teach pendant and keeps the enable switch in “gently pressed” state to reset the emergency stop state and turn on the servo power.

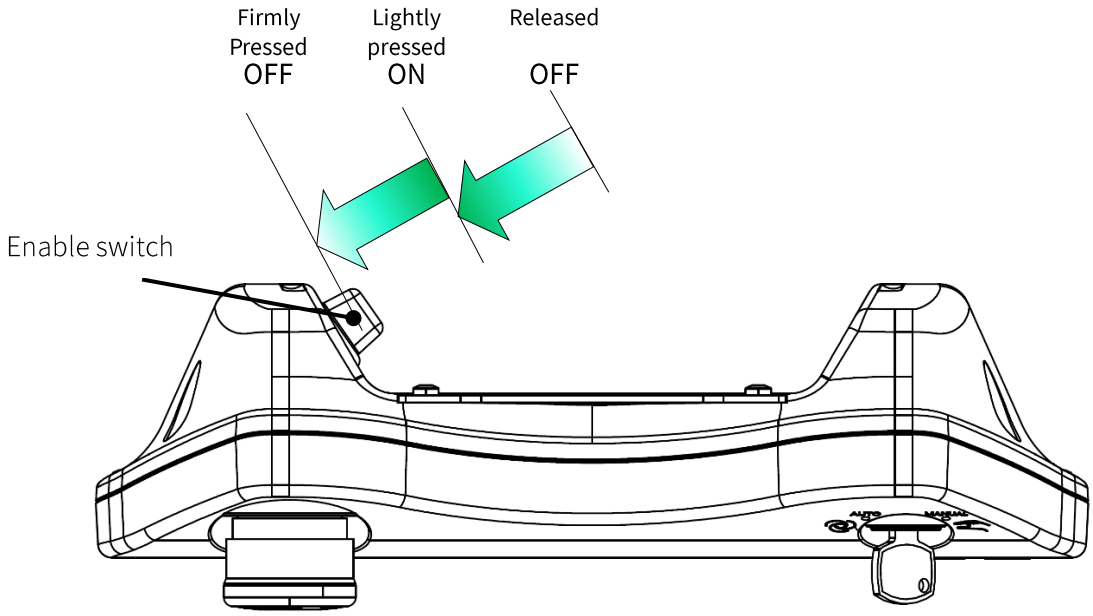
The enable switch is of the three-position type; the switch stops the robot immediately and turns off the robot’s power source when it is “released” or “firmly pressed.”

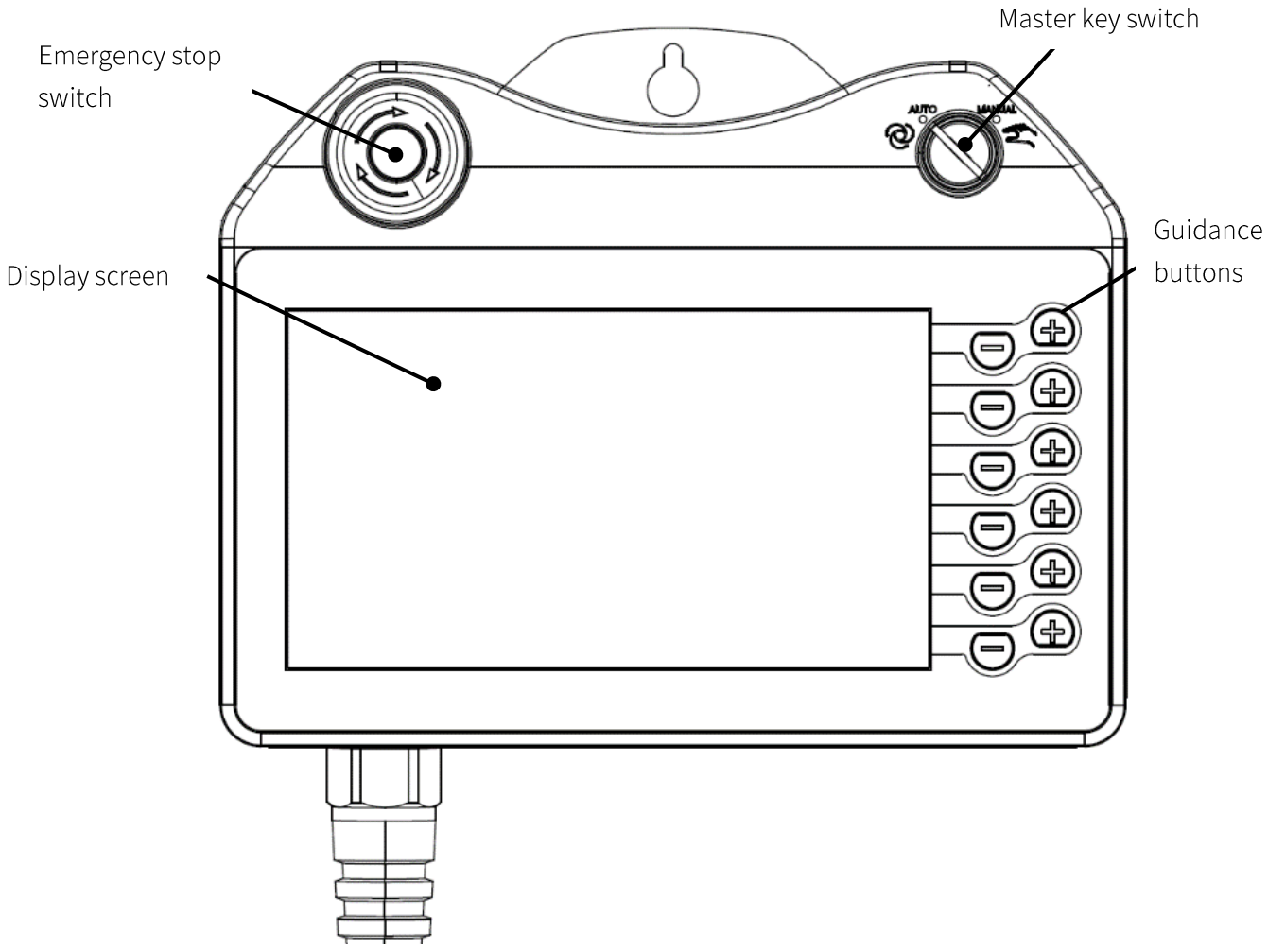
The servo cannot be turned on when the enable switch is OFF. Keep the enable switch ON when guiding the robot with the teach pendant.

[For TP1000]



[For TP5000]

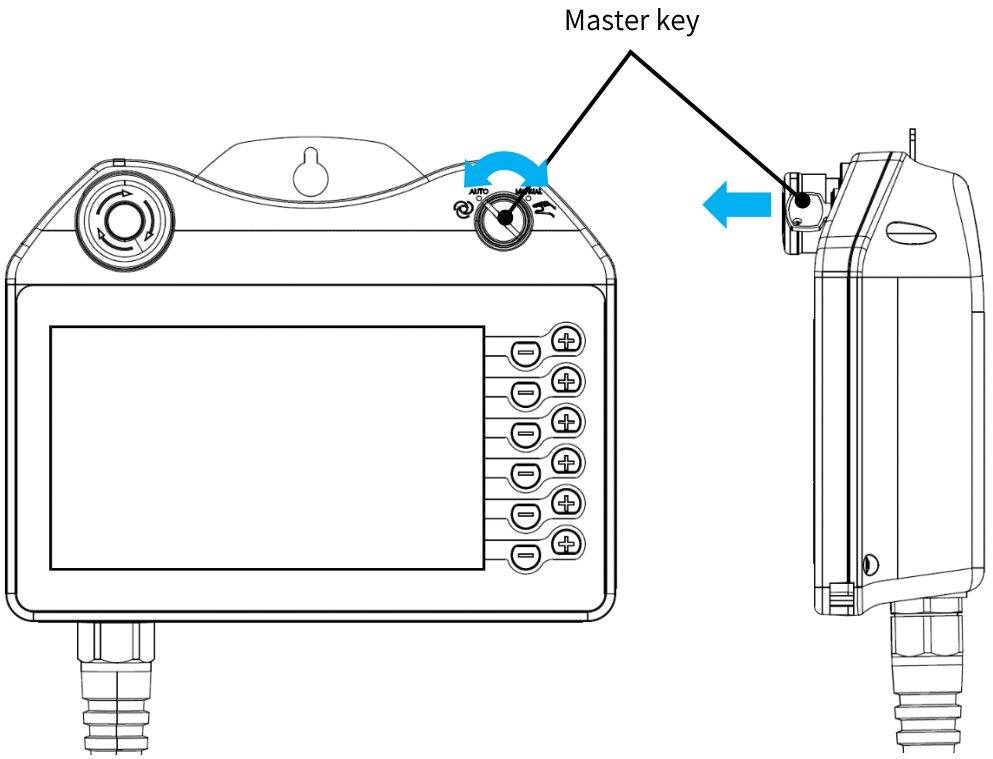

Firmly
Pressed
OFF
Lightly
pressed
ON
Released
OFF
Enable switch


Master key switch
Emergency stop
switch
Guidance
buttons
Display screen
AUTO
MANUAL

2.2.5 Master key switch

The master key switch is provided to select the MANUAL or AUTO mode of the control panel. When having to enter the working range in order to do teaching operation or inspection, the operator should select the TEACH mode and carry a key during the operation to prevent another operator from changing the mode.

> *MEMO*
>
> *When using the TP1000, since the master key switch is not at reach, be sure to work in the hazardous area with the teach pendant to be able to press the emergency stop button if necessary.*



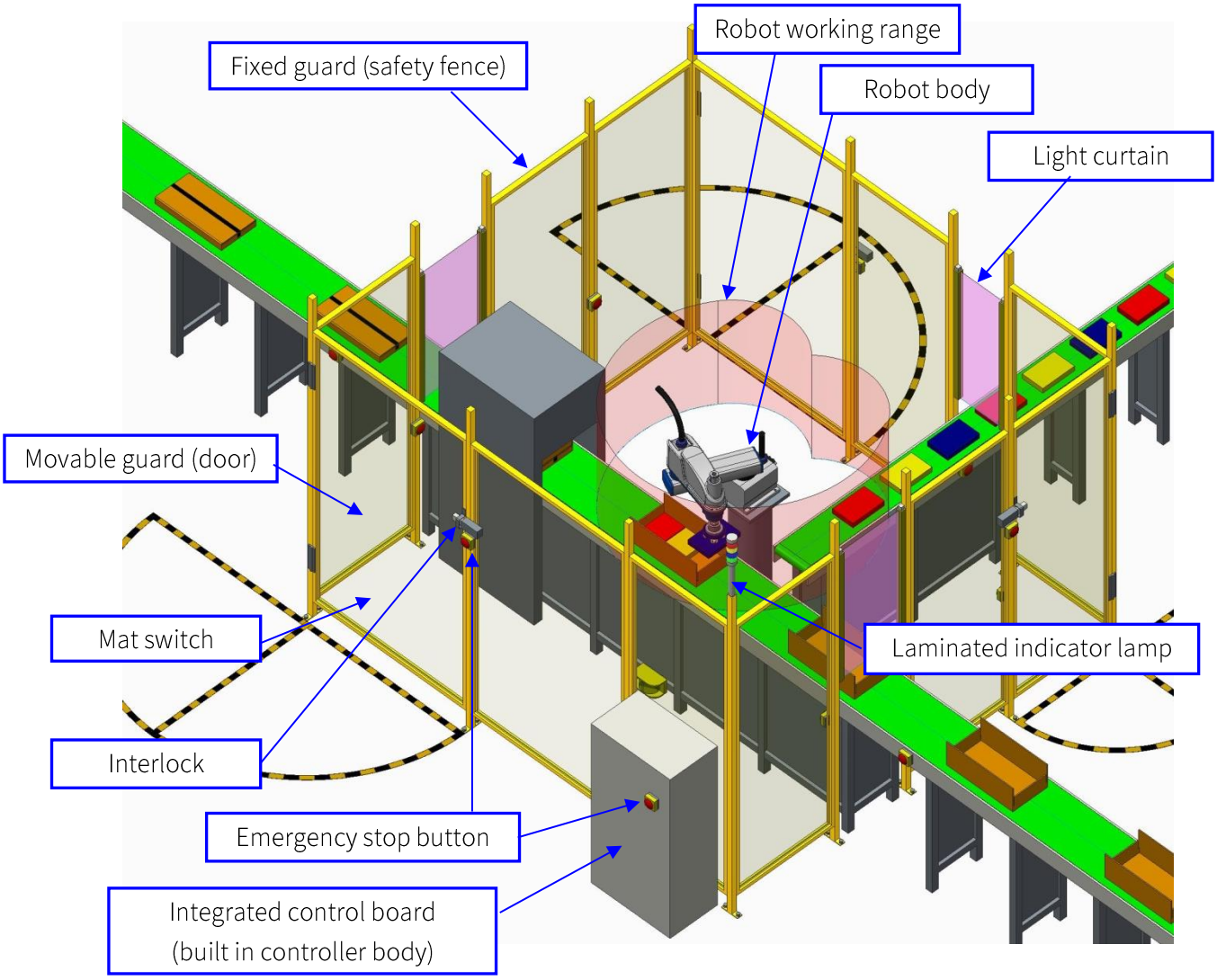
Front view of master key switch

Side view of master key switch

2.3 Examples of safety measures

Most of disasters caused by the robot originate from unsafe human behavior. When using the robot, the operator should foresee what will involve danger and try to prevent such a dangerous condition. Operation should be done only after all safety conditions are satisfied. Examples of main safety measures are shown below.

2.3.1 General cautions on using the robot



1) When installing the robot, provide a space required for performing the operation during installation and subsequent operations (teaching, operation, maintenance and disposal) safely.
   - ISO 13854: Minimum gaps to avoid crushing of parts of the human body

2) Identify the dangerous area, and install safety fences, etc. to prevent entry of any person. The dangerous area signifies an area near the robot, where a person who has entered it will jeopardize.
   - ISO14120: General requirements for the design and construction of fixed and movable guards
   - ISO13857: Safety distances to prevent hazard zone being reached by upper and lower limbs

3) Install an emergency stop switch at a location where it can be pressed immediately by an operator who has determined the occurrence of abnormality outside and inside the safety fence, etc., and connect it to the external emergency stop contacts (EMS1B to 1C and EMS2B to 2C) of the controller.
   - ISO13850: Emergency stop

4) Install interlock devices such as interlock switch, light curtain and mat switch at the entrance and exit doors of the safety fence and connect them to the safety contact input of the controller so that the robot stops if a person enters the dangerous area. The interlock devices should be electrically independent break-contact (normally-closed contact) devices.
   - ISO14119: Interlocking devices associated with guards
   - IEC61496-1,-2: Electro-sensitive protective equipment
   - ISO13855: Positioning of safeguards with respect to the approach speeds of parts of the human body
   - ISO13856-1: Pressure-sensitive protective devices

5) Install the controller at a safe location where an operator can overlook the operation of the robot.

6) To perform transportation, installation, teaching (teaching, program and operation check), maintenance (inspection, adjustment and repair) and disposal, be sure to observe the laws, rules and regulations defined in each country. Also, display the current condition of the robot to prevent an unrelated person from carelessly entering the working range and operating the robot.

7) Before daily operation, perform the following check.
   <Check before operation>
   - Visually confirm that the robot body, controller, peripheral equipment and cables are in good condition.
   - Make sure that no object stands in or new the working range of the robot and peripheral equipment.
   - Make sure that the emergency stop and other safety devices operate properly.
   - Make sure that no abnormal noise or vibration is involved in the robot operation.

8) Be sure to operate the pushbutton switches provided on the teach pendant while visually confirming them.

2.3.2 Cautions on operation

In principle, perform teaching and inspection outside the dangerous area of the robot. To perform them inevitably, strictly observe the following cautions.

1) Teaching should be performed after the drive source is shut off (servo OFF) whenever possible. The drive source should not be allowed to enter automatic state or to be recovered (servo ON) by other operator.
   Inspection should be performed after the primary power is shut off whenever possible. The primary power should not be allowed to enter automatic state or to be recovered by other operator.

2) The switches that change the operating state of the robot (start switch, stop switch, etc.) should not be allowed to be operated by a person other than the operator (by posting the display board, locking the operator panel, arranging watcher, etc.).

3) The following work regulations should be established so that the operation is performed in accordance with the regulations.
   ① Operation methods and procedures for starting the robot, handling the switches, teaching, inspection, and confirming them
   ② Operating speed suitable for situations (the speed is desirably made as slow as possible.)
   ③ Method for exchanging signals between operators
   ④ Recovery method for an abnormality
   ⑤ Method for resetting abnormal state and confirming safety before restarting the robot that has stopped abnormally
   ⑥ Other following methods for preventing the danger caused by unexpected movement of the robot and mis-operation
      • Methods for limiting operation of switches (start switch, stop switch, etc.) that select the operating state of the robot to an operator (by posting the display board, locking the operator panel, arranging watcher, etc.)
      • Method for allowing an operator in the dangerous area to hold the emergency stop device (by using the teach pendant)
      • Position and posture in which safety of an operator can be secured
      • Types and determination method of abnormalities (method for confirming alarm, etc.)
      • Method for preventing and reducing noise
      • Method for changing setup and confirming it (program change procedure, teaching data to be change, method for confirming it, etc.)
      • Method for exchanging signals between robot operator and related equipment operator

   * The work regulation should try to be prepared by inviting suggestions about the description suitable for respective sites from concerned operators, facility design engineers, etc.

4) The operations should always be performed by two persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other.

5) The operator should do the job in an attitude ready to press the emergency stop pushbutton switch at any time. Also, the operator should perform the job at a position from which the operator can evacuate immediately at the time of an emergency after confirming the robot working range and shields in the surroundings. Also, the operator should be careful so as not to turn operator's back to the robot during operation.

6) The supervisor should keep on the job at a position where the supervisor can overlook the entire robot system and operate the emergency stop pushbutton switch at the time of an emergency. Also, the supervisor should keep anyone from entering the dangerous area.

2.3.3 Other cautions

1) The gripping unit of the robot should not stick out, except for the part required for operation. Also, even at a sudden stop due to power failure, malfunction or emergency stop during operation, the robot should hold a workpiece in a stable posture.

2) If the robot has malfunctions, turn off the power, remove the cause of the malfunction, maintain the peripheral equipment and completely restore the malfunctioned robot. Then, confirm its movement a low speed. Even if the robot has stopped, do not approach the dangerous area immediately.

3) Turn off the power when the robot is not in use.

MEMO

Because of space limitations, this document covers only important safety precautions for the robot, but not all general safety information. Before carrying out work, read safety documents compliant with the laws, regulations and standards in each country or receive instructions from the consultants who are familiar with the rules, regulations and standards.

Also, carry out risk assessment and take measures to reduce risk.

2.3.4 Recommended protector

Observe the standards, regulations and rules in each country to carry out risk assessment, and then consider and select use.

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	
Earplug	Protection part: Ears Use: Protect from noise*1.	

*1): For details about the noise level and the method for measuring it, refer to the Instruction Manual "THE Series Robot Manual: Specifications." SHIBAURA MACHINE recommends that the protectors be worn for the operation in the environment where a single robot produces generates noise at 85 db or more based on the Guideline for the Prevention of Noise-Induced Impairments (Labor Standards Bureau Issue No. 546) in accordance with the laws in Japan.